

```
text='Hello! This is dhanush'
text_lower=text.lower()
print(text_lower)
```

hello! this is dhanush

```
import re
text='Hello! This is a DHANUSHKA from CSE'
text_clean=re.sub(r'^a-z','',text)
print(text_clean)
```

ellohisisafrom

```
import re
text='Hello! This is a DHANUSHKA from CSE'
text_clean=re.sub(r'^a-z',' 2 ',text)
print(text_clean)
```

2 ello 2 2 2 his 2 is 2 a 2 2 2 2 2 2 2 2 2 2 2 from 2 2 2 2

```
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords', quiet=True)
stop_words = set(stopwords.words('english'))
text="I'm going to college today"
token=[]
for word in text.split():
    if word not in stop_words:
        token.append(word)
print(token)
```

["I'm", 'going', 'college', 'today']

```
import nltk
nltk.download('punkt_tab', quiet=True)
sentence="I don't love our collage"
basic=sentence.split()
print('Basic split:',basic)
from nltk.tokenize import word_tokenize
smart=word_tokenize(sentence.lower())
print('NLTK tokenize:',smart)
```

Basic split: ['I', "don't", 'love', 'our', 'collage']
NLTK tokenize: ['i', 'do', "n't", 'love', 'our', 'collage']

```
import nltk
from nltk.stem import PorterStemmer, WordNetLemmatizer
nltk.download('wordnet',quiet=True)
stemmer=PorterStemmer()
lemmatizer=WordNetLemmatizer()
words=['running','studies','better','caring','happily']
for word in words:
    stemmed_word=stemmer.stem(word)
    lemma=lemmatizer.lemmatize(word,pos='v')
    print(f'{word:12}stem={stemmed_word}lemma={lemma}')
```

running	stem=runlemma=run
studies	stem=studilemma=study
better	stem=betterlemma=better
caring	stem=carelemma=care
happily	stem=happililemma=happily

```
def preprocess(text):
    text = text.lower()
    text = re.sub(r'^a-z]', ' ', text)
    tokens = text.split()
    filtered_token = []
    for w in tokens:
        if w not in stop_words:
            filtered_token.append(w)
            lemmatized_token = []
    for w in filtered_token:
        lemmatized_token.append(
            lemmatizer.lemmatize(w)
        )
```

```
        return ' '.join(lemmatized_token)
    print(preprocess('I am running to the stores to buy some groceries'))
```

```
running store buy grocery
```

```
from sklearn.feature_extraction.text import CountVectorizer
emails=[
    'cricket match cricket match cricket',
    'free win money free win money win',
    'cricket match free win money cricket',
]
cv=CountVectorizer()
x=cv.fit_transform(emails).toarray()
print('Vocabulary:',cv.get_feature_names_out())
print('bow matrix:')
print(x)
```

```
Vocabulary: ['cricket' 'free' 'match' 'money' 'win']
bow matrix:
[[3 0 2 0 0]
 [0 2 0 2 3]
 [2 1 1 1 1]]
```

```
from sklearn.feature_extraction.text import TfidfVectorizer
emails=[
    'cricket match cricket match cricket',
    'free win money free win money win',
    'cricket match free win money cricket',
]
tfidf=TfidfVectorizer(sublinear_tf=True,max_df=0.95,min_df=1)
x_tfidf=tfidf.fit_transform(emails).toarray()
print('Feature names:',tfidf.get_feature_names_out())
print('TF-IDF matrix:(rounded):')
```

```
Feature names: ['cricket' 'free' 'match' 'money' 'win']
TF-IDF matrix:(rounded):
```